MIL-M-52738A
8 July 1981
SUPERSEDING
MIL-M-52738
20 December 1972

MILITARY SPECIFICATION

MAT, LANDING; STEEL NON-PERFORATED (M8A1)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of non-perforated steel landing mats.

1.2 Classification. The mats shall be of the following types as specified:

Type I - Full panel
Type II - Female half panel
Type III - Male half panel

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, Flat and Seals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5680

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MILITARY

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

(Copies of specifications, standards and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other government documents, drawings and publications. The following other government documents form a part of this specification to the extent specified herein.

DRAWINGS

ME

TA13215E2384	- Mat, Landing; Half, Female, Steel, Non-Perforated (M8A1).
TA13215E2385	- Mat, Landing; Half, Male, Steel, Non-Perforated (M8A1).
TA13215E2386	- Mat, Landing; Full, Steel, Non-Perforated (M8A1).

PUBLICATIONS

DRDME - P715-144	- QA Pamphlet, Care, Operation and Maintenance of Inspection Equipment for Mat, Landing; Steel, Non-perforated (M8A1).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and reference cited herein, the text of this specification shall take precedence.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN WELDING SOCIETY

A3.0	Definitions, Welding and Cutting.
A5.1	Carbon Steel Covered Arc-Welding Electrodes.
A5.18	Carbon Steel Electrodes for Gas Metal Arc Welding.
D1.1	Structural Welding Code.

(Application for copies should be addressed to the American Welding Society, 2501 N.W. 7th Street, Miami, FL 33125)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The landing mat shall be in accordance with the applicable drawings and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish no less than 800 square feet of mat consisting of 36 type I, four type II and four type III panels for examination and test within the time frame specified (see 6.2) to prove, prior to starting production that his production methods will produce landing mats that comply with the requirements of this specification. Examination and tests shall be those specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.2.1 Weld joint specimens. In addition to the preproduction model, typical weld joint specimens shall be furnished and tested (see 6.2) by the contractor prior to the time the preproduction model is furnished. Specimens shall be prepared using production tooling. Tests shall be those specified in section 4 and shall be subject to surveillance and approval by the Government. Acceptable specimens shall represent the acceptable weld quality and will be used as a weld acceptance standard during production.

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3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.10).

3.4 Welding electrodes. Welding electrodes shall be in accordance with American Welding Society (AWS) A5.1 or A5.18 as required by welding process.

3.5 Assembly. When tested as specified in 4.5.2.1 mats shall assemble with like mats, and locking lugs, sliding bolts, and retainer keys shall aline and fully engage.

3.6 Identification marking. Each type of landing mat shall be identified in accordance with MIL-STD-130 and as shown on the applicable drawings.

3.6.1 Stenciling. Each bundle of landing mat shall be stenciled with Gothic capitals and numerals not less than 1 inch high. Stenciling shall be applied in accordance with MIL-T-704.

3.6.1.1 Bundles of individual mat. Bundles of landing mat bundled as specified in 5.2.1.1 and 5.2.1.2 shall be stenciled with the following information:

National Stock Number
Nomenclature
Contract Number
Mfg Identification
US

3.6.1.2 Bundles of type I, II, and III mat. Bundles of type I, II and III landing mat bundled as specified in 5.2.1.3 and as shown in figure 2 shall be stenciled as follows:

Mat Set Landing
Steel Nonperforated
NSN 5680-00-782-5577
268.8 Sq Ft

3.7 Gages. Unless otherwise specified the landing mats shall be inspected for all characteristics shown on the drawings under QAPs, Major, AQL 1.0 percent defective using inspection equipment conforming to the Government inspection equipment design drawings, and as shown in Quality Assurance Pamphlet DRDME P715-144. The inspection equipment will be loaned by the Government (see 6.9) or shall be fabricated and furnished by the contractor as specified (see 6.2).

3.8 Type I. Type I landing mats shall be as shown on TA13215E2386, and as specified herein.

3.9 Type II. Type II landing mats shall be as shown on TA13215E2384, and as specified herein.

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3.10 Type III. Type III landing mats shall be as shown on TA13215E2385, and as specified herein.

3.11 Workmanship.

3.11.1 Metal fabrication. Steel used in the fabrication of the landing mats shall be free of kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Finished mats shall be free of kinks, twists and unspecified bends. Corners shall be square and true. Flame cutting may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. All bends shall be made with metal dies or fixtures to insure uniformity of size and shape. All burrs, sharp edges and sharp corners resulting from the cutting and punching operation shall be removed by grinding, filing, abrasive rolling or blasting. All cutting and punching operations shall be from the top or wear surface of the mat.

3.11.2 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Structural Welding Code (D1.1)
Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Operators of any automatic, semi-automatic, or manual arc welding machines shall be considered welders for the purpose of this specification.

3.11.3 Welding. Steel welding shall be done by the shielded metal or inert gas shielded metal arc method. Projection welding may be used on the retainer key only. Defects shall be non-cumulative for each welding bead. Weld beads shall be regular and well formed.

3.11.3.1 Cleanliness. The surfaces of all parts to be welded shall be free from rust, scale, paint and other foreign matter. Before welding over previously deposited weld metal, all traces of slag shall be removed from the deposit by chipping and, if necessary, the deposit and adjoining base metal shall be wire brushed until clean at all points.

3.11.3.2 Cracks. There shall be no longitudinal, transverse or crater cracks. A crack is defined as a separation in the weld nugget or the heat effected zone. Craters shall be filled.

3.11.3.3 Weld size and length. Welds shall not be undersize nor more than 1/16 inch oversize for the size of weld specified. The cover plate bevel weld shall be continuous for the full length of the cover plate except that a blow back tolerance of 1/8 inch is acceptable. One 1/8 inch long unfused length of

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weld per bead is acceptable provided the defect is not within 3/4 inch of either end of the weld. A continuous fillet weld may be substituted in lieu of intermittent welds.

3.11.3.4 Undercutting and overlap. Undercutting and overlap shall not exceed 1/32 inch.

3.11.3.5 Porosity. A maximum of five each 1/16 inch diameter (maximum) pin holes or blow holes per cover plate is acceptable provided that none are closer than 1/2 inch and no more than three occur in any 1-1/2 inches of weld length. No burn through is acceptable.

3.11.3.6 Weld reinforcement. Weld reinforcement shall not exceed 1/8 inch.

3.11.3.7 Weld spatter. There shall be no sharp or pointed welds or weld spatter and traces of slag and weld spatter shall be removed prior to treatment, cleaning and painting.

3.11.4 Weld strength. The sliding bolt cover plate weld shall show no evidence of parting or fracture when tested as specified in 4.5.2.2.

3.11.5 Sliding bolt tack weld. Sliding bolts shall be tack welded in the open position prior to final treatment and painting.

3.11.6 Connector hook edge. The leading edge of the panels adjacent to the connector hooks shall have all burrs, sharp corners and sharp edges removed. If panel is formed by shearing or stamping, this operation shall be performed from the top side.

3.11.7 Welding definitions. Welding definitions are as specified in American Welding Society Specification AWS A3.0.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.1.2 Parts and components. Parts and components detailed on drawings shall be inspected in accordance with the Quality Assurance Provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring inspection, the sampling plan and the basis for acceptance and rejection (see 6.6 and 6.7.1).

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. Preproduction landing mat shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. Preproduction landing mat shall be tested as specified in 4.5.2. Failure of any test shall be cause for examining and preproduction panels in detail for compliance with this specification and reference drawings in regard to materials, dimensions, tolerances and workmanship. Landing mats not complying with such requirements shall be rejected.

4.4 Quality conformance inspection.

4.4.1 Unit of product. Individual type I, II or III landing mat shall be considered a unit of product.

4.4.2 Sampling.

4.4.2.1 Examination. Sampling for examination shall be in accordance with MIL-STD-105, inspection level S-4 except that the minimum sample size shall be 21 type I, four type II, and four type III landing mats and the maximum sample size shall not exceed one day's production.

4.4.2.2 Tests. Twenty type I, four type II and four type III landing mats shall be selected at random from each 1000 landing mats being produced from production tooling.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined for defects marked "X" in column 2 of table 1. AQL shall be 0.40 for critical defects, 1.5 for major defects, and 4.0 for minor defects.

TABLE I. Examination schedule.

Preproduction	Quality Conformance	Defect	Requirement Paragraph
1	2	3	4
		CRITICAL	
X	X	1. Burr, sharp corner or sharp edge on connector hook edge of the panel.	3.11.6

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TABLE I. Examination schedule. (Cont'd)

Preproduction 1	Quality Conformance 2	Defect 3		Requirement Paragraph 4
			MAJOR	
X	X	101.	Any part or component not in accordance with the QAP requirements as shown on the drawing.	3.1
X	X	102.	Any drawing requirements, other than those identified by QAP data, not as specified.	3.1
X	X	103.	Material not as specified.	3.3
X	X	104.	Welding electrodes not as specified.	3.4
X	X	105.	Identification marking not as specified.	3.6
X	X	106.	Bends not made with metal dies or fixtures.	3.11.1
X	X	107.	Welder's certification not as specified.	3.11.2
X	X	108.	Weld or weld method not as specified.	3.11.3
X	X	109.	All traces of slag of previously deposited weld not removed prior to additional welding.	3.11.3.1
X	X	110.	Weld contains cracks, pits, porosity, blow hole, blister, overlap or craters greater than allowed.	3.11.3.2 3.11.3.4 3.11.3.5
X	X	111.	Weld size and length not as specified.	3.11.3.3
X	X	112.	Sliding bolt not tack welded as specified.	3.11.5
			Minor	
X	X	201.	Flame cut surface contains ash or cooling check.	3.11.1
X	X	202.	Weld slag or spatter not removed prior to cleaning or painting.	3.11.3.7

4.4.4 Tests. Samples selected in accordance with 4.4.2.2 shall be tested as specified in 4.5.2.1 and 4.5.2.2. Failure of any test shall be cause for rejection.

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4.5 Inspection procedure.

4.5.1 Examination. Examination shall be in accordance with table I.

4.5.2 Tests.

4.5.2.1 Preproduction inspection test. Type I, II, and III landing mat shall be assembled as shown in figure 1. Evidence of binding and misalinement that prevents the locking lugs or sliding bolts from operating as specified shall constitute failure of this test.

4.5.2.2 Weld strength test. With the panel in the faceup position, a straight bar of sufficient size and strength to resist any visible bending due to the applied load shall be inserted 1 inch under the open end of each cover plate. The ends of the panel shall be restrained to prevent movement. A downward force of 135 pounds shall be applied to the bar 30 inches from the fulcrum which is located at the end of the panel. Evidence of parting or fracture of a weld shall constitute failure of this test (see 3.11.4).

4.6 Inspection of packaging.

4.6.1 Preproduction pack inspection.

4.6.1.1 Examination. Examine the preproduction pack for the defects specified in 4.6.2.3. Presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of pack.

4.6.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

112. Materials not as specified.
113. Arrangement of mats in bundle not as specified.
114. Strapping not zinc coated, properly tensioned, and sealed.
115. Strapping not positioned as specified in figure 2.
116. Marking illegible, incorrect, or incomplete.

5. PACKAGING

5.1 Preproduction pack. Unless otherwise specified (see 6.2) the contractor shall furnish a preproduction pack for examination and test within the time frame specified to prove, prior to starting production packing, that the applied

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packing and marking comply with the packaging requirements of this specification (see 6.4). Examination and tests shall be as specified herein.

5.2 Packing. Packing shall be level A (see 6.5).

5.2.1 Level A.

5.2.1.1 Type I. The full landing mat shall be bundled as specified herein. Each type I bundle shall contain 14 full panels nested together in pairs, bottom to bottom, in such a manner that the connector hooks will be protected between the rolled edge and first rib of the opposite panel. The two panels shall be staggered lengthwise approximately 3-1/4 inches to provide the minimum stacking height. The nested panel pairs shall be stacked to form a bundle. Strapping shall consist of five girthwise and one lengthwise straps located as shown in figure 2. Strapping shall conform to QQ-S-781, type I or IV, class B, 1.25 inch by 0.035 inch.

5.2.1.2 Type II and III. The female and male landing mats shall be bundled as specified herein. Type II and type III landing mats shall be bundled in a manner similar to that used for type I. Strapping shall consist of three girthwise and one lengthwise straps. Type II and type III shall be in separate bundles of 14 panels.

5.2.1.3 Type I, II and III. The landing mat panels shall be bundled as shown in figure 2 and as specified herein. Each bundle shall contain 13 type I, one type II and one type III panel. The panels shall be bundled as type I with the two half panels placed end to end and grouped with a full panel. The grouping containing the half panels shall be placed in the center of the stack. Strapping shall be as shown in figure 2.

5.3 Marking. Bundles shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The landing mats are intended for use as temporary airfield runway, taxiway and apron surfacing.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of mats required (see 1.2).
(c) Time frame required for submission of the preproduction model (see 3.2).
(d) Time frame required for submission of the weld joint specimens (see 3.2.1).
(e) Whether Government designed inspection equipment will be loaned by the Government or shall be fabricated and furnished by the contractor (see 3.7 and 6.8).

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(f) When preproduction pack is not required (see 5.1).
(g) Time frame required for submission of the preproduction pack when applicable (see 5.1).

6.3 Preproduction model. Any changes or deviations of production landing mats from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish landing mats, conforming to this specification.

6.4 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his oligation to pack and mark the landing mats in accordance with this specification.

6.5 Packing. Levels B and C packing have not been included, as the nature of the item is such that a lesser degree of packing would be unacceptable.

6.6 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include as a minimum, lot size, sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.7 Definitions.

6.7.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.8 Government inspection equipment design. Government inspection equipment design drawings exist and are listed on the top assembly. Gages conforming to the drawings should be used to validate the characteristics designated on the drawings.

6.9 Government-loaned property. Unless otherwise specified, the following property in the quantities indicated will be loaned by the government as specified in 3.7. The contracting officer should arrange to loan, when applicable, the inspection equipment:

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Item No.	Description	Quantity
1	Inspection equipment for Mat, Landing; steel non-perforated (M8A1)	1
2	QA pamphlet, care operation and maintenance of Inspection equipment for Mat, Landing; Steel, non-perforated (M8A1)	1

6.10 Recycled Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification. (See 3.3).

Custodians:
Army - ME
Navy - YD

Preparing activity:
Army - ME

Project 5680-0079

Review activity:
Navy - MC

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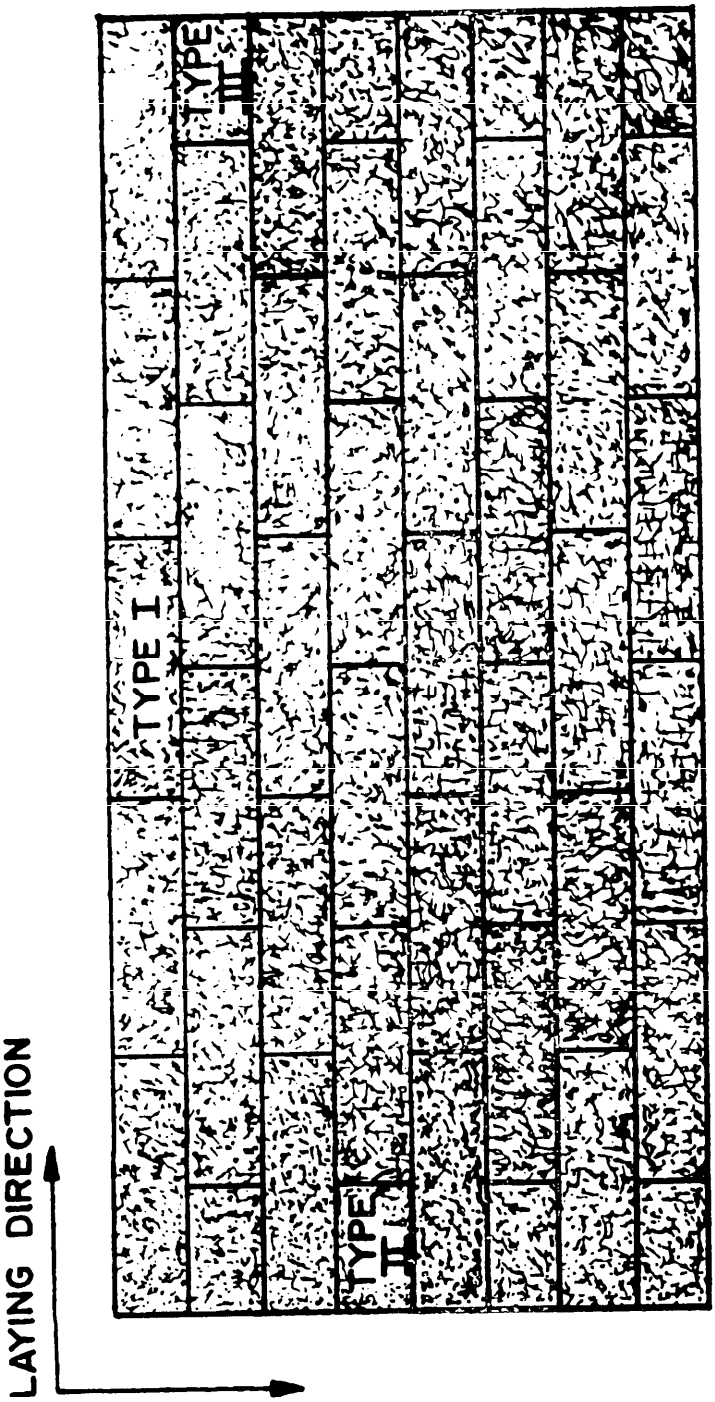


FIGURE 1. Preproduction mat assembly test pattern.

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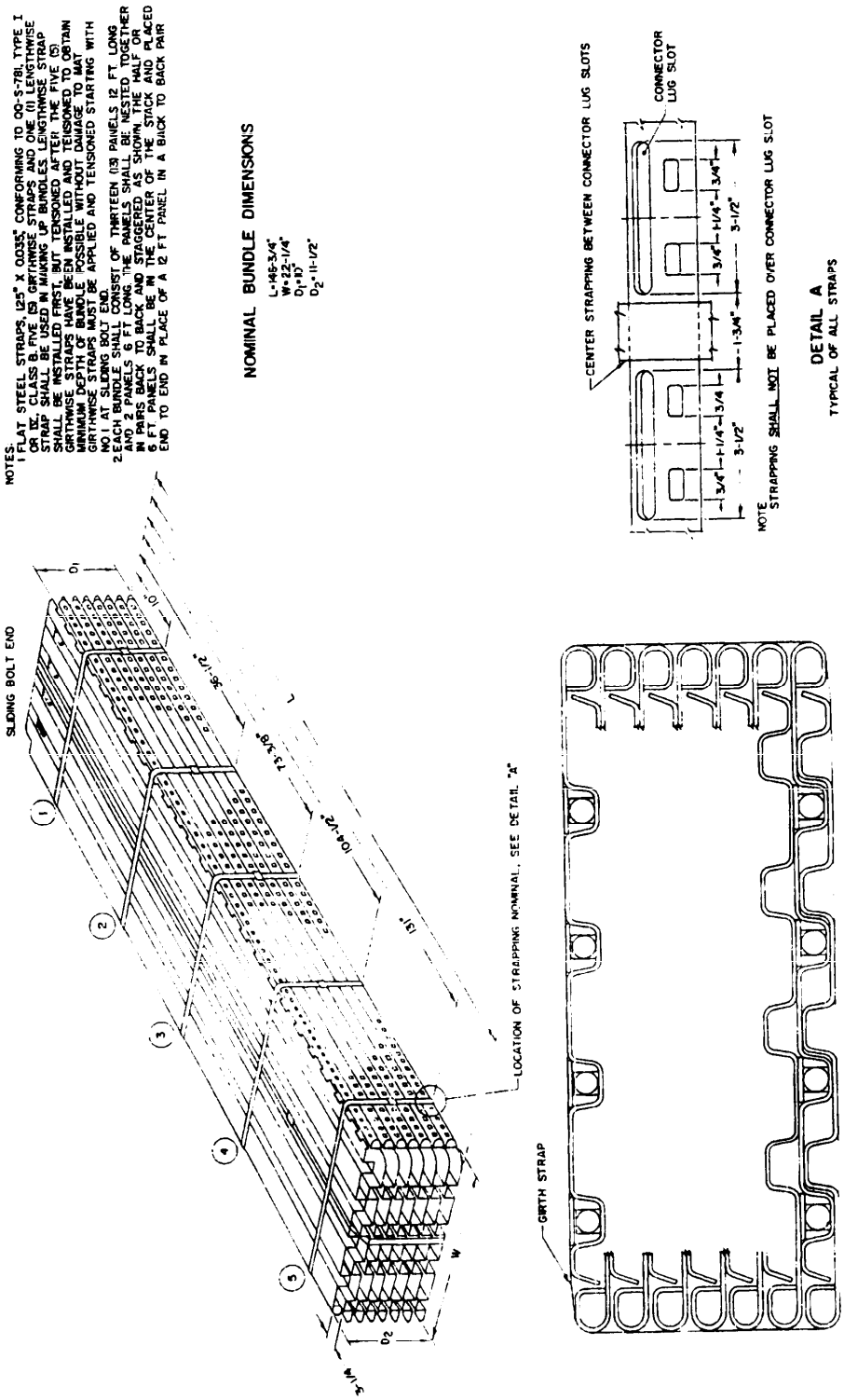


FIGURE 2. Level A packing.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-M-52738A Mat, Landing; Steel Non-Perforated (M8A1)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

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DATE

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